Machine learning
ITA0611

Assignment-1

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① Explain how Machine learning can identify Poverty-Prone regions. using socio-economic data

Identify Poverty-Prone regions, using Machine learning

SDG1 : No Poverty

Machine learning can identify areas where people are more likely to live in poverty by analyzing socio-economic data.

Steps :

* Collect data such as income, employment, education housing, healthcare, and population.
* Clean and prepare the data
* Use ML Algorithms, such as Decision Tree, Random Forest or K-means Clustering
* Train the model using historical poverty data
* Predict which regions have a high-medium or low poverty risk.
* Display the results using a map

Example: If a region has low income, high unemployment and poor education facilities, ML may classify it as a poverty-prone region

Benefit:

Government can identify needy areas and provide better employment, education, and financial support

② Design a Machine learning model to identify eligible beneficiaries for government welfare schemes.

Identify eligible beneficiaries for government welfare schemes - SDG1 - No Poverty

Machine learning can help governments select people who are eligible for welfare schemes based on their socio-economic conditions.

Steps:

* collect data such as income, family size, employment or disability, and housing and housing conditions
* Define eligibility rules using existing beneficiary records
* Train a classification algorithm such as Decision Tree or Random Forest
* Give new applicant data to the model
* The model predicts Eligible / NOT Eligible
* Government officials can verify the prediction before providing benefits

Example:

A low-income family with many dependents may be identified as eligible for a food or financial assistance scheme.

Benefits:

Faster identification of beneficiaries and reduction of errors and duplicate applications.

(3) Explain how Machine learning can predict crop yield using historical Agricultural data.

Machine learning can predict how much crop will be produced using historical Agricultural and weather data.

Steps:

* collect previous data about rainfall, temperature, soil type, fertilizer, irrigation, area, and crop type.
* clean and preprocess the data
* use algorithms such as linear Regression, Random Forest, or XGBoost.
* Train the model using previous crop-yield records
* Enter current Agricultural conditions
* The model predicts the expected crop yield

Example:

Based on rainfall, soil quality, and fertilizer usage the model can predict that a farm may produce 3 tonnes of rice per hectare.

Benefits:

Farmers can plan irrigation, fertilizer usage and harvesting, helping improve food production.

(4) Develop a machine learning solution for crop disease detection using leaf images

Machine learning can detect crop diseases by analyzing images of plant leaves.

Steps:

* collect images of healthy and diseased leaves.
* label the images according to the disease
* preprocess and resize the images.
* Train an image classification model such as CNN.
* Give a new leaf image to the model.
* The model identifies whether the leaf is healthy or diseased and can predict the disease type.

Example:
A farmer uploads a tomato leaf image. The model detects early blight.

Benefit: Early disease detection helps farmers take action quickly, reducing crop loss, and increasing food production.

(5) Explain how Machine learning can reduce food wastage in supermarkets.
Machine learning can help supermarkets predict how much food will be sold and reduces excess stock.

Steps: *collect historical sales data
*Include factors, such as season, holidays, discounts weather and customer demand
*Train a prediction model such as linear Regression Random Forest or XGBoost.
* Products close to expiry can be offered at discounts or redirected to food-donation program.

Example: If Machine learning that only 100 milks Packets will be sold tommorrow the supermaket can avoid purchasing 200 Packets.

Benefit:
Reduces food waste, financial loss, and unnecessary production, supporting SDG2-Zero Hunger.